

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/377435467>

Automated vehicles for older adults with cognitive impairment: a survey study

Article in *Ergonomics* · January 2024

DOI: 10.1080/00140139.2024.2302020

CITATION

1

READS

154

7 authors, including:



[Junho Park](#)

The University of Calgary

25 PUBLICATIONS 116 CITATIONS

[SEE PROFILE](#)



[Maryam Zahabi](#)

Texas A&M University

77 PUBLICATIONS 1,057 CITATIONS

[SEE PROFILE](#)



[Marcia G Ory](#)

Texas A&M University

671 PUBLICATIONS 24,184 CITATIONS

[SEE PROFILE](#)



[Me Benden](#)

Texas A&M University

93 PUBLICATIONS 993 CITATIONS

[SEE PROFILE](#)

Automated vehicles for older adults with cognitive impairment: a survey study

Junho Park, Maryam Zahabi, Xi Zheng, Marcia Ory, Mark Benden, Anthony D. McDonald & Wei Li

To cite this article: Junho Park, Maryam Zahabi, Xi Zheng, Marcia Ory, Mark Benden, Anthony D. McDonald & Wei Li (16 Jan 2024): Automated vehicles for older adults with cognitive impairment: a survey study, *Ergonomics*, DOI: [10.1080/00140139.2024.2302020](https://doi.org/10.1080/00140139.2024.2302020)

To link to this article: <https://doi.org/10.1080/00140139.2024.2302020>



Published online: 16 Jan 2024.



Submit your article to this journal [↗](#)



View related articles [↗](#)



View Crossmark data [↗](#)

ARTICLE



Automated vehicles for older adults with cognitive impairment: a survey study

Junho Park^a, Maryam Zahabi^b, Xi Zheng^c, Marcia Ory^d, Mark Benden^d, Anthony D. McDonald^e and Wei Li^f

^aDepartment of General Engineering, Santa Clara University, Santa Clara, CA, USA; ^bWm Michael Barnes '64 Department of Industrial and Systems Engineering, Texas A&M University, College Station, TX, USA; ^cWyze, Kirkland, WA, USA; ^dSchool of Public Health, Texas A&M University, College Station, TX, USA; ^eDepartment of Industrial and Systems Engineering, University of Wisconsin-Madison, Madison, WI, USA; ^fDepartment of Landscape Architecture and Urban Planning, School of Architecture, Texas A&M University, College Station, TX, USA

ABSTRACT

As the population is ageing, the number of older adults with cognitive impairment (CI) is increasing. Automated vehicles (AVs) can improve independence and enhance the mobility of these individuals. This study aimed to: (1) understand the perception of older adults (with and without CI) and stakeholders providing services and supports regarding care and transportation about AVs, and (2) suggest potential solutions to improve the perception of AVs for older adults with mild or moderate CI. A survey was conducted with 435 older adults with and without CI and 188 stakeholders (e.g. caregivers). The results were analysed using partial least square – structural equation modelling and multiple correspondence analysis. The findings suggested relationships between older adults' level of cognitive impairment, mobility, knowledge of AVs, and perception of AVs. The results provided recommendations to improve older adults' perception of AVs including education and adaptive driving simulation-based training.

Practitioner summary: This study investigated the perception of older adults and other stakeholders regarding AVs. The findings suggested relationships between older adults' level of cognitive impairment, mobility, knowledge of AVs, and perception of AVs. The results provided guidelines to improve older adults' perception of AVs.

ARTICLE HISTORY

Received 27 July 2023
Accepted 2 January 2024

KEYWORDS

Automation; transportation; ageing; driving; independence

1. Introduction

The population of Americans aged 65 and older is projected to grow from 58 million in 2021 to 88 million by 2050 (He, Goodkind, and Kowal 2016; Monica, Díaz-Santos, and Vossel 2021; Nash 2019). About 10–20% of older adults in the U.S. live with mild or moderate cognitive impairment (CI) and the prevalence of mobility issues among these individuals is high (Pedersen et al. 2014; Stats 2006). Individuals with mild CI exhibit either amnesic CI (memory problems) or nonamnesic CI (reasoning or speed of processing problems) (O'Connor et al. 2010). People with moderate CI experience difficulties in multiple cognitive domains and in performing activities of daily living (ADLs) (Kempf et al. 2016; O'Connor et al. 2010). CI also involves decrements in complex functional tasks such as everyday memory, planning, and organisation (Farias et al. 2006).

Transportation and continued community mobility play a crucial role in the lives of older adults with

cognitive impairment (Thomas and Gutches 2020). Maintaining mobility and preventing disability are key to living independently. As individuals age, they may experience changes in their mobility due to changes in gait, balance, and physical strength. These changes can increase the number and severity of falls, making it harder for older adults to visit friends and family and continue performing their activities independently (Webber, Porter, and Menec 2010). Older adults who lose their mobility are less likely to remain living at home, have higher rates of disease, disability, and hospitalisation, and a poorer quality of life compared to those who have not lost their mobility (Grimmer et al. 2019). Therefore, developing ways to prevent or reverse loss of mobility is critical to help older adults maintain independence. This is not only a matter of physical health but also the social and emotional well-being of older adults. Planning for mobility changes in the years to come is important for older adults who want to stay independent in their own homes and communities.

1.1. Autonomous vehicles

Automated vehicles (AVs) in this study refer to the highest level of automation based on the Society of Automotive Engineers (SAE) levels of automation (i.e. Level 5) (SAE 2021) in which the driving system performs all the dynamic tasks of driving in all conditions and the driver is not expected to take control of the vehicle. However, the driver can interact with the in-vehicle information system to perform different tasks (e.g. changing the destination or turning on music). Prior studies suggested AVs as a viable solution to improve mobility of older adults with cognitive impairment (Knoefel et al. 2019). However, there are issues with human-system interactions in AVs (Lee and See 2004; Parasuraman and Riley 1997). One of the major limitations of AVs is the level of driver's engagement and trust. Some studies argued that AVs still cause passive fatigue for drivers, although they might reduce active fatigue (May and Baldwin 2009; Saxby et al. 2013). Active fatigue in AVs is attributed to the driver's engagement in supervisory tasks, such as monitoring the vehicle's automation and being ready to take control if needed. This cognitive workload and sustained attention to the driving task can lead to fatigue. Passive fatigue, on the other hand, occurs when the AV is in control, and the driver has less to do. This reduced engagement and stimulation may lead to complacency and decreased alertness, resulting in passive fatigue. When using public transportation, older adults are not engaged in supervisory or monitoring tasks (since they are passengers and not drivers) and therefore, using these transportation options might not require high level of mental resources for older adults. However, using public transportation can be a cognitively demanding task for older adults with CI, especially when travelling to new places. Furthermore, there could be physical accessibility issues with public transportation (Ravensbergen et al. 2021, 2022). For example, there are still concerns with having low-floor vehicles, accessible bus stops/train stations, adequate seating, and handholds, which could make older adults reluctant to use public transportation. These physical barriers also exist for personal AVs. For example, older adults and individuals with cognitive impairment may face challenges getting in and out of AVs due to mobility issues (e.g. high step-in height or narrow door openings) (Faber and van Lierop 2020; Knoefel et al. 2019; Maresova et al. 2018). In addition, physical interactions with AV interfaces, such as adjusting seats or controls, can be problematic for those with cognitive impairment (e.g. small or hard-to-reach buttons and controls,

complex or confusing interfaces, and lack of tactile feedback) (Park et al. 2023).

1.2. Older adults and autonomous vehicles

Cognitive impairment (CI) can negatively affect older adults' mobility (M). About 16 million people in the U.S. with CI cannot drive or stop driving due to the limited capability of cognitive processing speed, memory, attention, or decision making (Eskandar et al. 2022; Perneczky et al. 2006; Ranchet et al. 2017; Zahabi et al. 2017). Compared with drivers without CI, cognitively impaired older drivers are less likely to drive and rate driving conditions as more difficult (Lyman, McGwin, and Sims 2001). Older adults with mild cognitive impairment (MCI) have reduced driving performance such as poor lane change and speed control (Wadley et al. 2009). Therefore, they may face with driving restrictions or quitting driving, which results in reduced access to social activities (Stanley et al. 2011) and relying on family members or caregivers for a ride (Bascom and Christensen 2017). The spatial range of mobility of elderly people with CI is restricted, with most out-of-home time spent close to their residences (Shoval et al. 2011). Although public transit can offer a viable alternative for some people, many older adults lack access to public transit and rely on friends and family for transportation (Mittra, Bae, and Ritchie 2019). However, this dependence on others can cause scheduling difficulties and feelings of indebtedness, which might decrease the quality of life (Metz 2000).

Previous studies found that demographic characteristics such as age or gender can affect older adults' ability to access transportation services and, therefore, their mobility (Litman 2021). Ignoring the demographic differences in mobility needs and patterns can result in transportation systems that are not accessible or convenient for certain groups of people. Older adults with mobility impairment might have difficulty walking long distances or climbing stairs to access public transportation (Litman 2021). There is evidence suggesting that the relationship between mobility and cognition in older adults is moderated by age, with the associations generally increasing with age (Demnitz et al. 2018). For example, lower extremity mobility impairment was significantly associated with poorer global cognitive performance, especially in older adults (Tolea and Galvin 2016). Therefore, older adults with mobility impairment might have difficulty walking long distances or climbing stairs to access public transportation. Women may have different

travel patterns, or mobility, than men due to household responsibilities or safety concerns (Litman 2021). Previous studies have found gender differences in cognitive functioning and decline account for different activity levels among men and women (Okamoto et al. 2021).

Other studies found that cognitive reserve, which is defined as the capacity to delay age-related CI, plays a significant role in how older adults understand and interact with AVs (Wöbbeking-Sánchez et al. 2020). The greater the cognitive reserve, the lower the CI presented. This suggests that older adults with a higher cognitive reserve may have a better understanding and interaction with AVs. However, even adoption of partially automated vehicles might present challenges for older adults with declining cognitive abilities (Knoefel et al. 2019). This indicates that the degree of CI could potentially influence older adults' perception towards AVs and understanding of AVs.

The relationship between mobility (M) and knowledge of AVs (KAV) has been investigated briefly in some studies. It was found that individuals who can use public transportation or ride-sharing services due to the degraded mobility, may be more open to the idea of AVs, as they are already accustomed to being driven by someone else (Ravensbergen et al. 2021, 2022). The comfort and familiarity of being a passenger, coupled with the potential benefits of AVs such as increased safety and convenience, could make these individuals more accepting of this technology.

Knowledge of AV could have an impact on perception of AV (PAV). Duncan et al. (2015)'s study suggested that even a short 15-minute presentation on AVs' capabilities (KAV), potential benefits, and safety features significantly improved older adults' opinion of and willingness to use AVs. A report from National Research Council (2014) notes that people with limited mobility, such as those with disabilities or the elderly, may benefit greatly from AVs, as these vehicles have the potential to increase their mobility and independence. However, the report also notes that older individuals with limited mobility due to the cognitive impairment may need additional education and training to improve their knowledge of AVs and to be able to fully understand and utilise AVs. Therefore, the level of cognitive impairment and age can potentially moderate the relationship between mobility and knowledge of AV.

In this study, we assessed perception of older adults regarding AVs in terms of: (1) their interest in using AVs (Rahman et al. 2019; Siegfried et al. 2021),

(2) whether they believe AVs can improve their independence (Knoefel et al. 2019), and (3) whether AVs can help them navigate better. Whether older adults are interested in using AVs depends on their level of trust in the capabilities of autonomous technology to operate safely and consistently (Edwards 2020; Huang et al. 2021). Many older drivers report concerns that automated driving technologies could fail to recognise and deal with all conditions on the roads, particularly in emergency situations (Zandieh and Acheampong 2021). This can potentially influence the duration and quality of the travel with AVs. Furthermore, studies of Charness et al. (2018) and Classen et al. (2021) found older adults' prior knowledge of AVs was significantly associated with less concern in using these vehicles, willingness to relinquish driving control, and better perception of AV.

1.3. Stakeholders' perception of AVs for older adults

A few studies focused on the opinions of stakeholders regarding older adults' perception of AVs. Stakeholders believe AVs can be beneficial for the mobility of older adults with cognitive impairment (Fraade-Blanar et al. 2021). One study focused on five categories of stakeholders including users (e.g. elderly people), legislators (e.g. government), operators, associations (e.g. car-sharing associations), and manufacturers (Hamadneh, Duleba, and Esztergár-Kiss 2022). Another study grouped the caregivers as informal (those who are not paid to give care) and formal (those who are paid to give care) caregivers (van Boekel et al. 2019). Stakeholders who worked closely with older adults with cognitive impairment or AVs emphasised the importance of the usability of AVs and older adults' capabilities (e.g. memory or decision making) to learn to use the technology (van Boekel et al. 2019). Privacy, confidentiality, and the timing of the introduction of the technology were also important factors that could affect older adults' use of AVs. Caregivers give preference to older adults' safety more than autonomy. This means that they could perceive the merits of AVs for those individuals negatively if the safety is not guaranteed (Landau et al. 2010).

1.4. Research gaps and objectives

Although some prior studies investigated the factors that can affect older adults' perception of AVs, these investigations were mainly focused on older adults without CI and only investigated relationships between

two or a subset of variables. In addition, research on how to improve the perception of older adults with cognitive impairment regarding AVs was limited. Therefore, the objectives of this study were to: (1) understand the perception of older adults and stakeholders providing services and supports regarding care and transportation about AVs, and (2) suggest potential solutions to improve the perception of AVs for older adults especially those with mild or moderate cognitive impairment (MMCI) as individuals with severe dementia need continuous human assistance and cannot use AVs alone.

2. Method

2.1. Participants

An online survey was completed by 435 older adults (without CI ($n = 123$), mild CI ($n = 229$), or moderate CI ($n = 83$)) and 188 stakeholders including individuals who work in transportation industry ($n = 44$, e.g. vehicle manufacturing companies or federal highway administration), family caregivers or representatives of healthcare ($n = 97$), and social service organisations ($n = 47$) that provide services or support older adults with CI. The sample size for both older adults and stakeholders satisfied the conservative sample size requirement for structural equation models (SEM), which required at least 120 data points (Wolf et al. 2013). A detailed description of participants' demographic information is provided in the results section.

2.2. Survey questions and hypotheses for older adults

Five hypotheses (H) were formulated in this study to assess the relationships between older adults' level of CI, mobility (M), knowledge of AV (KAV), and perception of AV (PAV). The list of hypotheses and references are summarised in Table 1.

To test these hypotheses, a survey composed of 11 questions (including two demographic questions) was distributed among older adults (above 65 yrs. of age).

The question types included: (1) yes/no questions with space for elaboration, (2) Likert scale response questions with ranges of responses between 1 and 5, or (3) checkbox questions. The questions were designed to understand older adults' perception of AVs and asked for information regarding physical and cognitive health and familiarity with AVs. At the beginning of the survey, an AV was clearly defined as a vehicle that is capable of sensing its environment and moving safely with little or no human inputs.

The indicators/questions of each latent variable were designed based on previous studies and are described in Table 2. For example, having difficulty in walking (Q5 in Table 2) was used to reflect any mobility issues because spatial and temporal movement ability was previously used as a measure of older adults' mobility (Soubra, Chkeir, and Novella 2019). Older adults who can walk short distances may rely on their cars for transportation when driving is an option (Viljanen et al. 2016). On the other hand, those who have difficulty in walking may need to prioritise accessibility and may rely on alternative forms of transportation if they are no longer able to drive. Vehicle access (Q4) and driver's licence status (Q3) were also used in prior studies as indicators of older adults' mobility (Fraade-Blanar et al. 2021; Shen et al. 2021). Older adults who have a driver's licence are more likely to own a car or have previously owned one (Hassan et al. 2019; Haustein and Siren 2014). Having a licence allows them to legally operate a vehicle, making car ownership more practical. Conversely, if older adults no longer have a valid driver's licence due to age-related factors or health concerns, they may need to give up car ownership, impacting their mobility. The combination of these factors significantly affects the mobility of older adults (Haustein and Siren 2014). Owning a car and having a driver's licence enhances mobility and independence, while the loss of either can restrict mobility, potentially leading to a greater reliance on others or public transportation.

Level of familiarity with AVs (Q7) and access to advanced driver-assistance systems (ADAS) (Souders,

Table 1. A list of hypotheses for older adults.

Hypothesis ID	Description	References
H_1	Level of cognitive impairment (CI) significantly and negatively affects mobility (M)	Eskandar et al. (2022); Pernecky et al. (2006); Ranchet et al. (2017); Zahabi et al. (2017)
H_2	Level of older adults' mobility (M) significantly and positively affects knowledge of AV (KAV)	Ravensbergen et al. (2021); National Research Council (2014)
H_3	Knowledge of AV (KAV) significantly and positively affects perception of AV (PAV)	Charness et al. (2018); Duncan et al. (2015); Edwards (2020); Huang et al. (2021)
H_4	Level of cognitive impairment (CI) significantly and negatively affects knowledge of AV (KAV)	Wöbbecking-Sánchez et al. (2020); Knoefel et al. (2019)
H_5	Level of cognitive impairment (CI) significantly and negatively affects perception of AV (PAV)	Lajunen and Sullman (2021); Knoefel et al. (2019)

Table 2. Survey questions for older adults.

Question ID	Latent variable	Question	Response type
Q1	Demographic information (DI)	Your current age in years is	Free response
Q2		Your gender is	Multiple choice
Q3	Mobility (M)	Your current driving status is	Multiple choice
Q4		Your current vehicle access is	Multiple choice
Q5		How difficult is it for you to walk for a quarter of a mile (about 2–3 city blocks or for about 5–10 minutes)?	Multiple choice
Q6	Cognitive impairment (CI)	How would you describe your cognition—that is, thinking skills needed to engage in everyday activities?	Multiple choice
Q7	Knowledge of AV (KAV)	How familiar are you with autonomous vehicles? (A vehicle that is capable of sensing its environment and moving safely with little or no human inputs)	Multiple choice
Q8		Do you own or have access to a vehicle with advanced driver-assistance systems (e.g. adaptive cruise control, lane keep assist)?	Yes/No
Q9	Perception of AV (PAV)	How interested would you be in using automated vehicles (if affordable)?	Multiple choice
Q10		Do you think autonomous cars will help older adults stay independent longer?	Multiple choice
Q11		Do you think autonomous cars will help those with memory problems navigate their environment better?	Multiple choice

Note: Options for each question are listed in Table 5.

Best, and Charness 2017) (Q8) were used to indicate older adults' knowledge of AVs as previous studies indicated a significant correlation between familiarity and knowledge of new technology (Charness et al. 2018). Perception of AVs was measured with the degree of interests in using AVs (Q9), and effect of AVs on older adults' independence (Q10) and navigation (Q11). Q9 is a commonly used measure of perception of autonomous vehicles in research studies. Several previous studies measured interests in using AVs or willingness to use AV as a measure of perception of AVs (Rahman et al. 2019; Siegfried et al. 2021). Having interest in using AVs was associated with positive attitudes towards AV and trust in the technology (Choi and Ji 2015). Demographic information (DI) was asked to assess their relationship with mobility (Q1 and Q2 in Table 2). Prior studies found that older age was associated with lower levels of physical activity and mobility (Sallis, Owen, and Fisher 2015). Another study found that older age was associated with a greater risk of mobility disability, defined as difficulty walking a quarter of a mile or climbing a flight of stairs (Guralnik et al. 1995). Gender was also found to have a significant influence on mobility due to the physical and social characteristics differences between males and females (St-Onge 2005; Valentine et al. 2009).

CI was measured based on subjective reports on how older adults would describe their cognition with multiple choices including: no CI, mild CI, moderate CI, and severe CI. Three respondents identified as having severe CI and their data were not included in this

analysis. Self-reported measures are effective methods for detecting mild cognitive impairment and dementia based on prior studies (Kasai et al. 2021). During the informed consent process, participants were asked if they understood what they were being asked to do in this study. If they answered 'Yes' to this question, the online survey was presented to them. All participants were able to access the survey using their personal computer or laptops. Table 2 provides the details of survey questions for older adults. Based on these hypotheses and survey questions, an initial model was developed (Figure 1).

2.3. Survey questions and hypotheses for stakeholders

Six additional hypotheses (H') were formulated to explore the relationships between stakeholders' occupation and their thoughts on older adults' perception of AVs and mobility (Table 3). In general, we expected that stakeholders who work closely with older adults (i.e. primary care providers, organisations that provide services to older adults) would have positive opinion on older adults' perception of AVs (van Boekel et al. 2019) while stakeholders who work in transportation industry would have moderate opinion on older adults' perception of AVs.

The list of survey questions for stakeholders is provided in Table 4. The first question was asked to determine the industry/occupation of the stakeholders. The second question was asked to see if they have experience working directly with older adults with

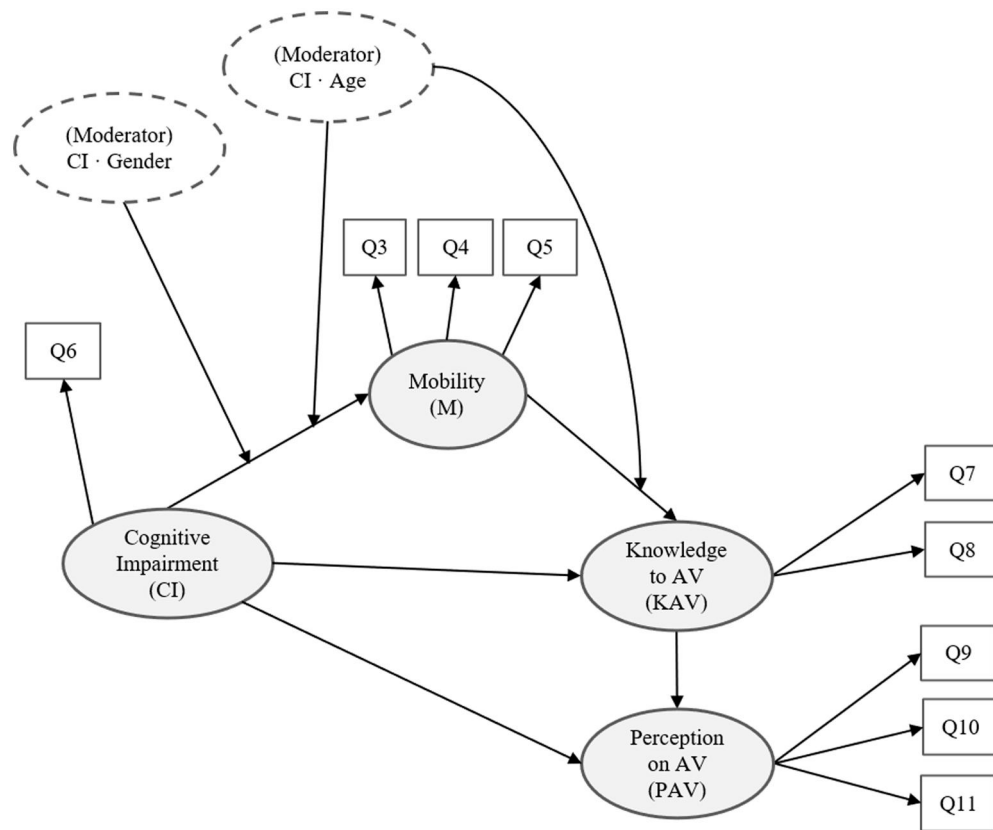


Figure 1. Initial model.

Table 3. Hypothesis testing result for stakeholders' survey data.

Hypothesis ID	Description	References
H'_1	Primary care providers would have positive opinion regarding older adults' interest in AVs.	van Boekel et al. (2019)
H'_2	Stakeholders from organisations that provide services or supports to older adults would have positive opinion regarding older adults' interest in AVs.	Classen et al. (2011); Kacperski, Vogel, and Kutzner (2020); Landau et al. (2010)
H'_3	Stakeholders from transportation industry would have moderate opinion regarding older adults' interest in AVs.	Landau et al. (2010)
H'_4	Primary care providers think AVs can significantly improve older adults' mobility.	van Boekel et al. (2019)
H'_5	Stakeholders from organisations that provide services or supports to older adults think AVs can significantly improve older adults' mobility.	Classen et al. (2011); Kacperski, Vogel, and Kutzner (2020); Landau et al. (2010)
H'_6	Stakeholders from transportation industry think AVs may affect older adults' mobility.	Landau et al. (2010)

MMCI as having direct experience might impact their opinion regarding the use of AVs for this population. Questions 4, 5, and 6 are related to the questions asked in older adults' survey (questions 9, 10, and 11) to compare these responses from different perspectives. Question 7 was asked to provide recommendations on how to improve AVs in future to fit the needs of older adults with CI. Participants could select multiple responses for questions 4 and 7.

2.4. Survey procedure

The survey was advertised via social media channels (e.g. LinkedIn and Facebook) and was administered using a web-based survey platform (Qualtrics). Participants were first asked to fill out an online consent form and a demographic questionnaire before completing the survey. Responses were collected and organised using Qualtrics-online service between 3 March 2021, and 25 April 2021. The study protocol

Table 4. Survey questions for stakeholders.

Question ID	Question	Response type
Q ₁	Please tell us a little about yourself	Multiple choice
Q ₂	Do you work directly with older adults with mild to moderate cognitive impairment?	Yes/No
Q ₃	How familiar are you with autonomous vehicles? (A vehicle that is capable of sensing its environment and moving safely with little or no human inputs)	Multiple choice
Q ₄	What do you see as the major advantages for someone with mild or moderate cognitive impairment using autonomous vehicles?	Multiple choice
Q ₅	How interested do you think people with mild and moderate cognitive impairments would be in using automated vehicles (if affordable)?	Multiple choice
Q ₆	Do you think autonomous vehicles would improve people's mobility (i.e. the ability to get around the city freely and quickly) with mild to moderate cognitive impairment?	Multiple choice
Q ₇	What are the major barriers to using autonomous vehicles for someone with mild or moderate cognitive impairment?	Multiple choice

*Note: Options for each question are listed in Table 6.

was approved by Texas A&M university institutional review board (IRB) (No. IRB2020-1015D).

2.5. Data analysis

Two analyses were conducted for each of the survey results from older adults and stakeholders including partial least square - structural equation modelling (PLS-SEM) and multiple correspondence analysis (MCA). Responses to multiple choice questions were rated using the scales of 1–3, 1–4, or 1–5 depending on the number of options provided.

PLS-SEM was used to understand the structure and relationship of the latent variables (Hair et al. 2017). Instead of using regression analysis which enables one-way causation and can only handle observed variables (i.e. indicators or survey questions), we used structural equation modelling (SEM) because it is designed to handle both latent (unobserved) and observed variables. Furthermore, SEM allows for development of a complex path model with direct and indirect effects. PLS-SEM is a more exploratory and predictive modelling technique compared to covariance-based SEM, which is commonly used (Hair and Alamer 2022; Kono and Sato 2022). We set all indicators to be reflective of each construct because the construct's items are postulated to reflect one specific concept (Hair and Alamer 2022) and indicators do not completely form the constructs (Coltman et al. 2008). For example, 'income' can be another indicator for the demographic information, however, we did not ask this question in our survey. To evaluate the model, indicator reliability, internal consistency, convergent validity, and discriminant validity were calculated (Hair et al., 2021). In addition, we assessed any collinearity issues, significance/relevance of the structural relationships, and model's explanatory power. Rstudio with

SEMinR package was used to conduct this analysis (Hair et al., 2021).

MCA was conducted to investigate the underlying structures in the dataset and to identify associations between the levels of categorical variables in stakeholders' responses (Le Roux and Rouanet 2010). MCA is free from assumptions and by working with categorical data, it has the capability to represent linear and non-linear relationships equally well (Rodriguez-Sabate et al. 2017). Dimensions identified during MCA were normalised between 0 and 1; a value lower than 0.1 indicates a poor representation of the variable in the computed dimensions. After identifying the dimensions with MCA, we conducted hierarchical clustering on principle components (HCPC) to test the hypotheses (Kassambara 2017). The validity of MCA was tested with eigenvalues and v-test (Rodriguez-Sabate et al. 2017). Eigenvalues represent the relative relevance of each dimension to the total inertia. Thus, dimensions with an eigenvalue lower than 0.05 were not considered. V-test was used to test whether the categorical variables are included in specific cluster with population (i.e. all responses) (Husson, Lê, and Pagès 2011). If the test statistics is larger than 2 or less than -2, the null hypothesis can be rejected. To conduct these analyses, Factoshiny R package was used (Vaissie, Monge, and Husson 2015).

3. Results

3.1. Descriptive information

In total, 435 older adults completed the survey and were included in our analyses (Age: $M = 68.65$ yrs., $SD = 3.90$ yrs., Min age = 65 yrs., Max age = 80 yrs.). More than half of participants did not have access to vehicles ($n = 241$, 55.40%) and experienced some mobility issues ($n = 329$, 75.63%). About 74% of participants had at least moderate familiarity with AVs,

Table 5. Summary of responses from older adults.

Variables and categories	Frequency of occurrence (count)	Percentage (%)	Variables and categories	Frequency of occurrence (count)	Percentage (%)
Gender			Familiarity with AV		
Female	136	31.26	Extremely	73	16.78
Male	299	68.74	Very	90	20.69
Driver's licence			Moderately	159	36.55
Currently have	243	55.86	Slightly	64	14.71
Previously had	185	42.53	Not at all	49	11.27
Never	7	1.61	Interested in AV		
Vehicle access			Definitely	161	37.03
Currently, own a car	176	40.46	Probably	134	30.80
Used to own (Not now)	226	51.95	Possibly	106	24.37
Can have access	18	4.14	Probably not	29	6.67
No access for the last 3 years	15	3.45	Definitely not	5	0.23
Difficulty to walk			Experience in advanced driver-assistant systems		
Not at all	106	24.37	Yes	244	56.09
A little difficult	150	34.48	No	191	43.91
Somewhat difficult	159	36.55	AV improves independence		
Very difficult	20	4.60	Definitely	141	32.41
Cognitive impairments			Probably	155	35.63
No	123	28.28	Possibly	124	28.51
Mild	229	52.64	Probably not	14	3.22
Moderate	83	19.08	Definitely not	1	0.23
			AV improves navigation		
			Definitely	132	30.34
			Probably	129	29.66
			Possibly	147	33.79
			Probably not	16	3.68
			Definitely not	11	2.53

mentioned that they were interested in AVs, and thought AVs would improve their independence and navigation. Descriptive statistics on survey responses are shown in Table 5.

In addition to older adults, 188 participants responded to the stakeholders' survey (Table 6). The stakeholders were individuals who work in transportation industry (e.g. vehicle manufacturing company, federal highway administration), family caregivers or representatives of healthcare, and social service organisations that provide services or support to older adults. Participants who selected the 'others' category for their occupation had the option to provide comments. Most comments were related to occupations in the human factors and ergonomics area. A majority of responders (>80%) indicated moderate to extreme familiarity with AVs and positively evaluated the effect of AVs on the mobility of older adults with CI (>70%). However, they had concerns regarding lack of control (17.41%), equipment failures (16.7%), and older adults' capability to use AVs (16.34%). After completing the survey, participants entered a drawing for a \$60 gift card.

3.2. Structural equation modeling for older adults' survey data

The standardised root mean square residual was 0.09 (Hu and Bentler 1998) and the normed fit index was

0.937 (Lohmöller 1989), which indicated a good model fit (Byrne 1998). The initial model was also assessed for reliability and validity assumptions of SEM. For example, the factor loading of Q4 under mobility variable was too small (<0.5), which indicated that the observed variables under mobility were not strongly related to the latent factor (mobility) they were intended to measure. Since the responses for Q3 and Q4 were highly correlated, we decided to remove Q4 from the model. Thus, after removing Q4, the PLS-SEM analysis was conducted again.

Figure 2 illustrates the modified and final model after ensuring that all the assumptions were met. The loading or path coefficient (β) means that if the predictor construct increases by one standard deviation unit, the endogenous construct will increase by β standard deviation unit ($\alpha = 0.05$). First, we conducted confirmatory factors analysis on the measurement model. In this model, the indicator reliability values for Q3, Q5, Q6, Q7, Q10, and Q11 were above 0.708 (Hair et al., 2021). The indicator reliability values of Q8 (0.687) and Q9 (0.701) were slightly lower than 0.708, however, researchers frequently obtain weaker indicator loadings (<0.708) for their models in social science studies especially when newly developed scales are used (Hulland 1999). Internal consistency was measured with Cronbach's alpha, rhoC, and rhoA, which passed the recommendation for exploratory research (i.e. minimum rhoC was 0.784) (Dijkstra 2014; Dijkstra

Table 6. Summary of variables of stakeholders.

Variables and categories	Frequency of occurrence (count)	Percentage (%)	Variables and categories	Frequency of occurrence (count)	Percentage (%)
Occupation			Older adults' interests in using AVs		
Primary care	78	41.49	Not interested	3	1.60
Healthcare organisation	47	25.00	Moderately interested	75	39.89
Transportation	44	23.40	Slightly interested	66	35.11
Others	19	10.11	Very interested	44	23.40
Directly work with older adults with cognitive impairment			Improved mobility with using AVs		
Yes	25	13.30	Yes	132	70.21
No	163	86.70	Maybe	49	26.06
Familiarity with AVs			No	7	3.72
Extremely familiar	17	9.04	Major barriers of using AVs for older adults with mild or moderate cognitive impairment (selected all that apply)		
Very familiar	47	25.00	Inability to use	92	16.34
Moderately familiar	90	47.87	Equipment failures	94	16.70
Slightly familiar	27	14.36	Legal liability	74	13.14
Not familiar at all	7	3.72	Hacking of the vehicle	29	5.15
Major advantages of AVs			Privacy breaches	58	10.30
Improve mobility	66	35.11	Interactions with other vehicles	85	15.10
Improve roadway safety	74	39.36	Lack of control	98	17.41
Improve Independence	48	25.53	Older adults' family members objection	33	5.86

and Henseler 2015). Construct validity assumption was met since all values were greater than 0.5 for all constructs (Hair et al., 2021). Each construct's squared variance within (AVE) was larger than the shared variance between constructs, which showed discriminant validity (Fornell and Larcker 1981; Hair et al., 2021). Collinearity was tested using the variance inflation factor (VIF). All scores were less than 5, which indicated that there were no collinearity issues. Significance and relevance of the structural model relationships were assessed (Aguirre-Urreta and Rönkkö 2018) and all paths were found to be significant. Although the explanatory power (r^2) for PAV was weak (i.e. 10%), prior studies argued that the value as low as 0.10 can be considered satisfactory (Raithel et al. 2012). Table 7 summarises the hypothesis test results.

3.3 Multiple correspondence analysis (MCA) for stakeholders' survey data

Six questions from Table 4 were included in this analysis as they were multiple choice questions (Q2, Q3, Q4, Q5, Q6, and Q7). Thirteen dimensions were identified with eigenvalues higher than 0.05, which were considered to explain the nature of the survey data (Rodriguez-Sabate et al. 2017). Based on the v-test results, statistically significant relationships between stakeholders' occupations and their thoughts on older adults' perception of AVs and mobility were identified from the MCA. The results of hypotheses tests are summarised in Table 8.

4. Discussion

4.1. Implications of major findings

Hypothesis 1 (H_1) stated that the level of cognitive impairment significantly and negatively affects mobility. This hypothesis was supported by the data. This finding is in line with the results of prior studies that suggested older adults with cognitive impairment cannot drive or stop driving (based on the responses to Q3: current driving status) due to the limited capability of cognitive processing speed, memory, attention, or decision making (Eskandar et al. 2022). Cognitive risk factors can include behavioural slowing (Salthouse 2010) (e.g. difficulty in walking), limited capability in selective visual attention (Baldock et al. 2007), and lack of ability to execute planned actions under time pressure (Stelmach and Nahom 1992). All these risk factors can limit older adults' mobility. Older adults with cognitive impairment might need other passengers or caregivers in the vehicle with them because

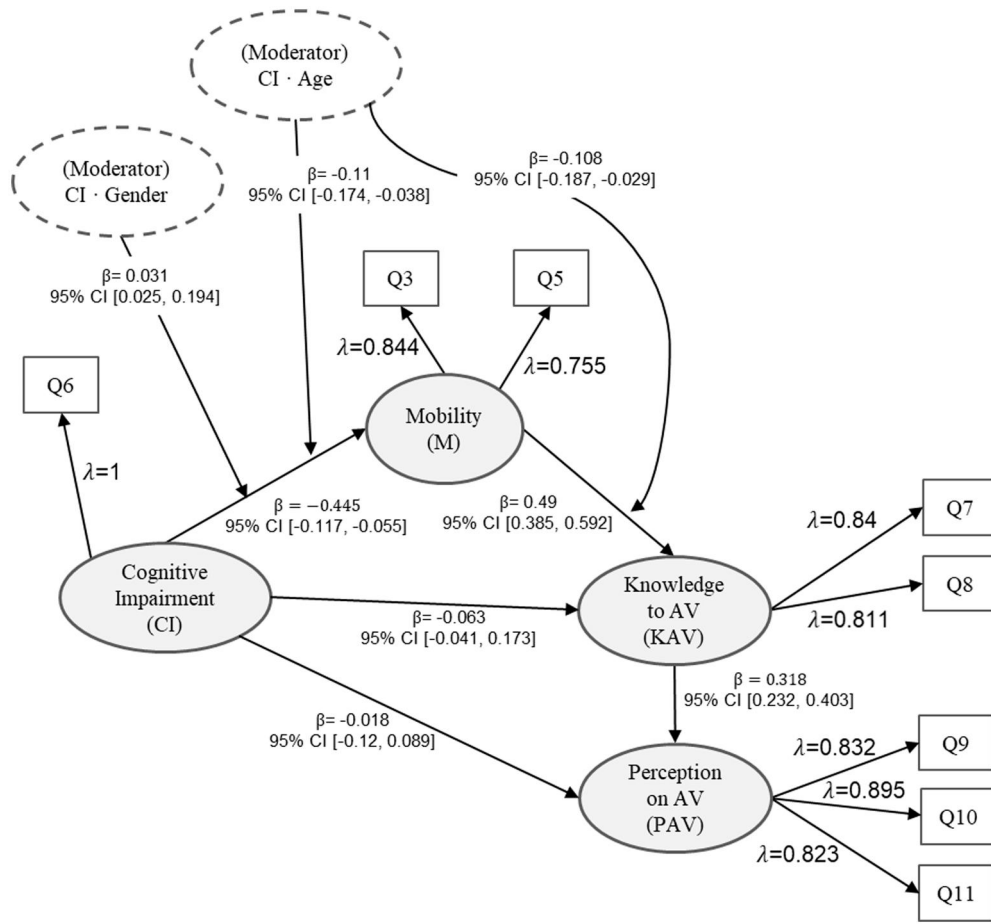


Figure 2. PLS-SEM for older adults.

Table 7. Hypothesis testing result for older adults' survey data.

Hypothesis ID	Description	Test result
H_1	Level of cognitive impairment (CI) significantly and negatively affects mobility (M).	Supported
H_2	Level of older adults' mobility (M) significantly and positively affects knowledge of AV (KAV)	Supported
H_3	Knowledge of AV (KAV) significantly and positively affects perception of AV (PAV)	Supported
H_4	Level of cognitive impairment (CI) significantly and negatively affects knowledge of AV (KAV)	Refuted
H_5	Level of cognitive impairment (CI) significantly and negatively affects perception of AV (PAV)	Refuted

Table 8. Hypothesis testing result for older stakeholders' survey data.

Hypothesis ID	Description	Test Result (p -value, v .test)
H'_1	Primary care providers would have positive opinion regarding older adults' interest in AVs.	Supported ($p = .62$, v .test = 1.12)
H'_2	Stakeholders from organisations that provide services or supports to older adults would have positive opinion regarding older adults' interest in AVs.	Supported ($p = .79$, v .test = 1.41)
H'_3	Stakeholders from transportation industry would have moderate opinion regarding older adults' interest in AVs.	Supported ($p = .84$, v .test = 1.45)
H'_4	Primary care providers think AVs can significantly improve older adults' mobility.	Refuted ($p < .0001$, v .test = 3.31)
H'_5	Stakeholders from organisations that provide services or supports to older adults think AVs can significantly improve older adults' mobility.	Supported ($p = .62$, v .test = 1.08)
H'_6	Stakeholders from transportation industry think AVs may affect older adults' mobility.	Supported ($p = .85$, v .test = 1.87)

they are concerned about memory loss and confusion in the middle of the road (Faber and van Lierop 2020). Therefore, there is a need to have customisable or easy-to-use in-vehicle supportive informational system for cognitively impaired older adults (Zahabi et al. 2021; Zahabi, Razak, et al. 2020).

Hypothesis 2 (H_2) stated that mobility significantly and positively affects knowledge of AV. This hypothesis was also supported by the data. The findings of previous studies partially support this finding. Satariano et al. (2012) highlighted the importance of mobility for older adults, suggesting that technologies

enhancing mobility could be of interest to older adults and can impact their usage among this population. Another possible explanation for this relationship might be that those who are more mobile may attend events or participate in activities that expose them to information about AVs. However, older adults with limited mobility might have fewer opportunities to encounter such information unless they actively seek it out through other means like online platforms or television.

Hypothesis 3 (H_3) stated that knowledge of AVs significantly and positively affects perception of AV. This hypothesis was supported by the data and the findings were similar to the results of prior studies that found knowledge of AV could positively influence perception on safe and consistent driving with AVs among older adults (Edwards 2020; Huang et al. 2021). A recent study also supported this finding in that previous knowledge could influence older consumers' perception towards AVs (Park and Han 2023). This is in line with Duncan et al. (2015)'s study which found that even a minimal amount of education on AVs significantly improved older adults' opinion of and willingness to use AVs. However, it is important to note that although this relationship was statistically significant, the explanation power was weak ($r^2 = 10\%$). Therefore, future studies should explore this relationship with additional variables or questions under each construct (Hair et al. 2021).

Proper education can improve older adults' knowledge of AVs. For example, older adults could learn that low levels of automated technology (Level 1 & 2 automation (SAE 2018)) can alleviate some of the transportation difficulties they faced previously by reducing the risks associated with ageing adults' most common crash scenarios (e.g. cross-traffic turns at intersections) (Duncan et al. 2015). This indicates the necessity to educate older adults not only on how to interact with AVs but also on safety, emergency handling, and consistency issues which are major concerns in using AVs. In fully automated vehicles, older adults need to have comfort without experiencing fear of equipment failure, which is also relevant to their perception of AVs (Duncan et al. 2015). Thus, vehicle safety instructions should be sufficiently explained to reduce any fear regarding equipment failure. Educational material can be conveyed in demonstration-based or video-based format. A previous study found that drivers who went through the demonstration-based training experienced lower workload when using low levels of automation while driving than those who were trained using the video-based

training method (Zahabi et al. 2021). Educational material can also be provided using virtual reality/augmented reality technologies which have been found to be effective training tools and are less expensive than driving simulation-based or on-road training (Lang et al. 2018).

The second potential solution to improve knowledge of AVs is adaptive driving simulation-based training (ADST) with customised in-vehicle interfaces (Zahabi, Park, et al. 2020) for older adults with cognitive impairment. ADST is a kind of training approach in which the problem, the stimulus, or task is varied as a function of how well the trainee performs (Kelley 1969). Adaptive training with a driving simulator might be a useful training medium for older adults with cognitive impairment as they indicated concerns of loss control and equipment failure in the survey. Traditional driving simulation approaches (i.e. non-adaptive solutions) might not be useful in training of older adults with MMCI due to the differences in learning styles and capabilities, memory capacity, level of impairment, individual preferences, or level of prior knowledge.

Driving simulation training has been used in several studies to acclimate older adults to the use of highly automated vehicles. For example, training programs that put older adults behind the wheel in a driving simulator, with an observer who helps them develop their skills, have been found to be beneficial in improving older adults' driving skills (Haeger et al. 2018). In another study, older drivers' trust and perceived safety increased after being exposed to a driving simulator operating in autonomous mode (Association for Psychological Science 2011). This suggests that driving simulator training can help increase trust and perceived safety among older adults. Furthermore, driving in a car simulator requires selective, sustained, distributed, and divided attention as well as executive abilities (Classen et al. 2020). These are cognitive skills known to decay in older age, and driving simulator training can help enhance these skills. Lastly, exposing older adults to an automated simulator may promote older adults' perception of AVs (Association for Psychological Science 2011).

Hypothesis 4 (H_4), which stated that cognitive impairment negatively affects knowledge of AVs, was not supported by the data. However, this insignificant relationship might have been due to the limited number of predictors under the CI and KAV variables (Knoefel et al. 2019; Wöbbeking-Sánchez et al. 2020). First of all, this online survey could not sufficiently categorise the responders' prior knowledge of AV. Older

adults who had more exposure to AV technology or were interested in it before the onset of cognitive impairment may retain some knowledge or awareness of AV. Secondly, the level of variations on the prior knowledge or education could have impacted their responses (Duncan et al. 2015; Souders and Charness 2016). Furthermore, the level of support and assistance from caregivers and family members could play a significant role in their responses, although they could not be captured in the online survey. If caregivers or family members educate and engage the older adults in discussions about AV, it may help maintain their knowledge to some extent. In addition, there could be differences in the level of access to information on AVs among older adults, which could have made variations in the responses (Souders and Charness 2016). Lastly, those with milder impairment may still be able to learn and retain some knowledge, while those with more severe impairment may have difficulty in grasping new concepts (Lajunen and Sullman 2021).

Hypothesis 5 (H_5), which stated that cognitive impairment negatively affects perception of AVs, was also not supported. The perception of AVs among older adults with cognitive impairment could have been influenced by factors like perceived usefulness and trust (Park and Han 2023). That is, positive attitudes are associated with a perception of these vehicles as useful and trustworthy. However, the predictors for the perception of AVs in this study included older adults' interest in using AVs, and their opinion regarding the effect of AVs on their independence and navigation. It is also possible that factors such as individual experiences or adaptability (or openness to new technologies) might have influenced older adults' responses.

Regarding stakeholder's survey data, hypothesis 1 (H'_1) stated that primary care providers would have positive opinion regarding older adults' interest in AVs. This hypothesis was supported by the data, which was similar to the findings of van Boekel et al. (2019). AVs are often equipped with advanced safety features, reducing the risk of accidents and improving the overall well-being of older adults, a matter of concern for primary care providers (Hassan et al. 2019). However, this does not mean that those older adults with cognitive impairment could be alone in AVs. Hypotheses 4 (H'_4), which stated that primary care providers think AVs can significantly improve older adults' mobility was refuted by the data. One possible reason for this could be safety issues that could happen if older adults are left alone in AVs. Other barriers include lack

of in-person (i.e. caregivers) support or cost to develop customised training methods for older adults (Landau et al. 2010; Solli et al. 2015). In addition, primary care providers' response to Q'_7 suggested that they do not think that older adults with cognitive impairment would be able to use AVs (Table 6), which might be due to concerns related to loss of control, equipment failures, and usability of AVs. These concerns could have negatively affected primary care providers' opinion (Haghzare et al. 2022).

Hypotheses 2 (H'_2) and 5 (H'_5) posited that organisations that provide services or supports to older adults would have positive opinion on older adults' perception of AVs and the effect of AVs on their mobility. These hypotheses were supported by the data. It is critical to understand the gap between the marketing presentations of AVs and their actual capabilities to be used for specific population (Dixon 2020), which is also called miscalibrated mental model of AV (Haghzare et al. 2022; Parasuraman and Riley 1997). We found that primary care providers who work directly with older adults with MMCI mentioned that older adults with MMCI would be moderately interested in using AVs and that AVs may improve their mobility. Primary care providers could narrow the gap in miscalibrated mental model because they work directly with older adults with cognitive impairment and know cognitive limitations of these individuals (Haghzare et al. 2022). However, stakeholders who work in organisations that provide services or supports to older adults may not directly work with older adults with cognitive impairment and therefore, responded more positively to the survey questions.

Two hypotheses (H'_3 and H'_6) posited that stakeholders in transportation industry would have moderate opinion on older adults' perception of AV and the effect of AVs on their mobility. These hypotheses were supported by the data. Stakeholders in transportation industry were concerned about AV technology-oriented issues such as equipment failure, taking over control, and legal liability (Duncan et al. 2015). AVs rely heavily on technology and equipment such as sensors, software, and hardware to operate. Any failure in these systems can lead to accidents or malfunctions, which could be particularly concerning for older adults who may not be as familiar with the technology or able to react quickly due to the cognitive impairment. Fully AVs are designed to operate without human intervention. However, there may be situations where older adults try to control the vehicle as they have cognitive impairment (e.g. memory loss). This could lead to hazardous situations. In the event of an

accident involving an AV, determining legal liability can be complicated, especially for older adults with cognitive impairment (legal issues or insurance claims). These factors can affect stakeholders' opinions regarding the interest of older adults in AVs and whether these technologies can impact their mobility. AV users need to be aware of the gap between the image of AVs in marketing and their actual capabilities and unforeseen risks of using these vehicles on road (Dixon 2020; Haghzare et al. 2022).

There could also be mismatches between the technology (AVs) reliability that was intended by stakeholders and older adults' perception of AVs. Some studies found that issues such as lack of reliability and difficulty of use can negatively affect older adults' perception of AVs (Liu, Zhang, and He 2019). Other studies emphasised the complexity of older adults' willingness to use AVs (Kadylak, Cotten, and Fennell 2021) and that factors such as AV reliability, can influence their acceptance or perception. While AVs offer technological solutions to enhance mobility of older adults, acceptance and perception among older adults play a critical role. This implies that the technology's reliability needs to align with the safety and mobility needs of older adults to bridge the gap in perception and acceptance (Doulabi, Hassan, and Li 2023).

4.2. Limitations and future work

The first limitation of this study was the period of data collection. The survey was distributed during the COVID-19 pandemic, and therefore, transportation mobility of people could be limited, especially for older adults. In addition, due to COVID-19, we used social media platforms to recruit participants, which could have biased the sample towards those individuals who have high digital literacy. Furthermore, there could be a selection bias in this study, which means the respondents were already able to use computer to answer the online survey (i.e. older adults with cognitive impairment who are less competent of using computers could have not joined this study).

Considering the online survey format and target population's cognitive limitation, we decided to minimise the number of questions, which led to having insufficient predictors under some latent variables. Second, the number of participants was not balanced based on gender or cognitive impairment level (i.e. about 70% of participants were males and 52.64% had MCI). Future studies might benefit from including a more balanced sample and a longer period of data collection. Third, due to the online survey platform,

level of cognitive impairment was measured using a self-reported (i.e. subjective) response, and therefore, it is not possible to tell if respondents were over or underestimating their cognitive abilities. Future studies might benefit from using in-person assessments such as the Montreal Cognitive Assessment (MoCA) screening tool (Nasreddine et al. 2005) to categorise individuals with cognitive impairment. Fourth, the mobility latent variable was measured with 'drivers' license', 'vehicle access' and 'ability to walk' and did not include factors such as 'access to public transportation'. Future studies could consider additional factors that could affect mobility of older adults. Also, since the study did not systematically measure older adults' acceptance, it is unclear how this factor could improve the perception of AVs among older adults with MMCI. Finally, the exploratory power of some relationships in PLS-SEM analysis was weak which might have been due to the limited number of questions in the survey.

The suggested educational/training programs have some benefits and limitations. ADST programs can provide hands-on experience and increase familiarity with AVs, which can enhance understanding and reduce anxiety or fear associated with new technology (Zahabi, Park, et al. 2020). It requires resources (i.e. purchasing driving simulators), can be time consuming, and therefore, may not be accessible to all older adults. Some older adults may find in-vehicle technology overwhelming or intimidating. Educational workshops can provide information about the benefits and operation of AVs and address any concerns (American Psychological Association 2017). However, they may not reach those who are homebound or have mobility issues.

5. Conclusions

The findings of this study suggested that the level of cognitive impairment significantly affects older adults' mobility. Older adults' mobility significantly and positively affected their knowledge of AVs. In addition, knowledge of AVs significantly and positively affects older adults' perception of AVs. Stakeholders who worked as primary care providers or in organisations that provided services to older adults with cognitive impairments responded positively on questions regarding older adults' interest in using AVs, while stakeholders who worked in transportation industries responded neutrally. Based on the findings, this study provided recommendations to improve older adults' perception and knowledge of AVs.

Acknowledgement

The opinions and conclusions expressed are solely those of the author(s) and do not represent the opinions or policy of Texas A&M.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research was funded by Texas A&M Triads for Transformation (T3).

References

- Aguirre-Urreta, M. I., and M. Rönkkö. 2018. "Statistical Inference with PLSc Using Bootstrap Confidence Intervals." *MIS Quarterly* 42 (3): 1001–1020. doi:10.25300/MISQ/2018/13587.
- American Psychological Association. 2017. "Cognitive Behavioral Therapy (CBT)." <https://www.apa.org/ptsd-guideline/treatments/cognitive-behavioral-therapy>
- Association for Psychological Science. 2011. "Driving Simulators Help Older Adults Improve Their Road Skills." Retrieved October 18 from <https://www.psychologicalscience.org/news/releases/driving-simulators-help-older-adults-improve-their-road-skills.html>.
- Baldock, M. R. J., J. Mathias, J. McLean, and A. Berndt. 2007. "Visual Attention as a Predictor of on-Road Driving Performance of Older Drivers." *Australian Journal of Psychology* 59 (3): 159–168. doi:10.1080/00049530701458035.
- Bascom, G. W., and K. M. Christensen. 2017. "The Impacts of Limited Transportation Access on Persons with Disabilities' Social Participation." *Journal of Transport & Health* 7: 227–234. doi:10.1016/j.jth.2017.10.002.
- Byrne, B. M. 1998. *Structural Equation Modeling with LISREL, PRELIS, and SIMPLIS: Basic Concepts, Applications, and Programming*. Psychology Press. doi: 10.4324/9780203774762
- Charness, N., J. S. Yoon, D. Souders, C. Stothart, and C. Yehner. 2018. "Predictors of Attitudes toward Autonomous Vehicles: The Roles of Age, Gender, Prior Knowledge, and Personality." *Frontiers in Psychology* 9: 2589. doi:10.3389/fpsyg.2018.02589.
- Choi, J. K., and Y. G. Ji. 2015. "Investigating the Importance of Trust on Adopting an Autonomous Vehicle." *International Journal of Human-Computer Interaction*, 31 (10): 692–702. doi:10.1080/10447318.2015.1070549.
- Classen, S., D. W. Eby, L. J. Molnar, B. M. Dobbs, and S. M. Winter. 2011. "Transportation and Ageing: Exploring Stakeholders' Perspectives on Advancing Safe Mobility." *South African Journal of Occupational Therapy* 41 (3): 18–24.
- Classen, S., J. Mason, S. W. Hwangbo, J. Wersal, J. Rogers, and V. Sisiopiku. 2021. "Older Drivers' Experience with Automated Vehicle Technology." *Journal of Transport & Health* 22: 101107. doi:10.1016/j.jth.2021.101107.
- Classen, S., J. Mason, J. Wersal, V. Sisiopiku, and J. Rogers. 2020. "Older Drivers' Experience with Automated Vehicle Technology: interim Analysis of a Demonstration Study." *Frontiers in Sustainable Cities* 2: 27. doi:10.3389/frsc.2020.00027.
- Coltman, T., T. M. Devinney, D. F. Midgley, and S. Venaik. 2008. "Formative versus Reflective Measurement Models: Two Applications of Formative Measurement." *Journal of Business Research* 61 (12): 1250–1262. doi:10.1016/j.jbusres.2008.01.013.
- Demnitz, N., D. B. Hogan, H. Dawes, H. Johansen-Berg, K. P. Ebmeier, M. J. Poulin, and C. E. Sexton. 2018. "Cognition and Mobility Show a Global Association in Middle-and Late-Adulthood: Analyses from the Canadian Longitudinal Study on Aging." *Gait & Posture* 64: 238–243. doi:10.1016/j.gaitpost.2018.06.116.
- Dijkstra, T. K. 2014. "PLS'Janus Face-Response to Professor Rigdon's 'Rethinking Partial Least Squares Modeling: In Praise of Simple Methods.'" *Long Range Planning* 47 (3): 146–153. doi:10.1016/j.lrp.2014.02.004.
- Dijkstra, T. K., and J. Henseler. 2015. "Consistent Partial Least Squares Path Modeling." *MIS Quarterly* 39 (2): 297–316. doi:10.25300/MISQ/2015/39.2.02.
- Dixon, L. 2020. "Autonowashing: The Greenwashing of Vehicle Automation." *Transportation Research Interdisciplinary Perspectives* 5: 100113. doi:10.1016/j.trip.2020.100113.
- Doulabi, S., H. M. Hassan, and B. Li. 2023. "Senior Americans' Perceptions, Attitudes, and Safety Concerns toward Autonomous Vehicles (AVs)." *Journal of Safety Research* 84: 218–231. doi:10.1016/j.jsr.2022.10.022.
- Duncan, M., N. Charness, T. Chapin, M. Horner, L. Stevens, A. Richard, D. Souders, J. Crute, A. Riemony, and D. Morgan. 2015. "Enhanced Mobility for Aging Populations Using Automated Vehicles" [Tech Report]. <https://rosap.nhtl.bts.gov/view/dot/29733>.
- Edwards, M. L. 2020. *Factors Influencing the Adoption of Autonomous Vehicles by the Elderly in Suburban Environments*. (Doctoral dissertation) Robert Morris University.
- Eskandar, M., W. C. Giang, S. Motamedi, H. Devos, L. Koon, A. E. Akinwuntan, and A. Kondyli. 2022. "Designing a Reminders System in Highly Automated Vehicles' Interfaces for Individuals with Mild Cognitive Impairment." *Frontiers in Future Transportation* 3. doi:10.3389/ffutr.2022.854553.
- Faber, K., and D. van Lierop. 2020. "How Will Older Adults Use Automated Vehicles? Assessing the Role of AVs in Overcoming Perceived Mobility Barriers." *Transportation Research Part A: Policy and Practice* 133: 353–363. doi:10.1016/j.tra.2020.01.022.
- Farias, S. T., D. Mungas, B. R. Reed, D. Harvey, D. Cahn-Weiner, and C. DeCarli. 2006. "MCI is Associated with Deficits in Everyday Functioning." *Alzheimer Disease and Associated Disorders* 20 (4): 217–223. doi:10.1097/01.wad.0000213849.51495.d9.
- Federal Interagency Forum on Aging-Related Statistics (US). 2008. *Older Americans 2008: Key Indicators of Well-Being*. Federal Interagency Forum on Aging Related Statistics.
- Fornell, C., and D. F. Larcker. 1981. "Evaluating Structural Equation Models with Unobservable Variables and Measurement Error." *Journal of Marketing Research*, 18 (1): 39–50. doi:10.1177/002224378101800104.

- Fraade-Blanar, L., N. Larco, R. Best, T. Swift, and M. S. Blumenthal. 2021. "Older Adults, New Mobility, and Automated Vehicles." University of Oregon.
- Grimmer, M., R. Riener, C. J. Walsh, and A. Seyfarth. 2019. "Mobility Related Physical and Functional Losses Due to Aging and Disease—a Motivation for Lower Limb Exoskeletons." *Journal of Neuroengineering and Rehabilitation* 16 (1): 2. doi:10.1186/s12984-018-0458-8.
- Guralnik, J. M., L. Ferrucci, E. M. Simonsick, M. E. Salive, and R. B. Wallace. 1995. "Lower-Extremity Function in Persons over the Age of 70 Years as a Predictor of Subsequent Disability." *The New England Journal of Medicine* 332 (9): 556–561. doi:10.1056/NEJM199503023320902.
- Haeger, M., O. Bock, D. Memmert, and S. Hüttermann. 2018. "Can Driving-Simulator Training Enhance Visual Attention, Cognition, and Physical Functioning in Older Adults?" *Journal of Aging Research* 2018: 7547631–7547639. doi:10.1155/2018/7547631.
- Haghzare, S., G. Delfi, E. Stasiulis, H. Mohamud, E. Dove, M. J. Rapoport, G. Naglie, A. Mihailidis, and J. L. Campos. 2022. "Can Automated Vehicles Be Useful to Persons Living with Dementia? The Perspectives of Care Partners of People Living with Dementia." *The Gerontologist* 62 (7): 1050–1062. doi:10.1093/geront/gnab174.
- Hair, J., and A. Alamer. 2022. "Partial Least Squares Structural Equation Modeling (PLS-SEM) in Second Language and Education Research: Guidelines Using an Applied Example." *Research Methods in Applied Linguistics* 1 (3): 100027. doi:10.1016/j.rmal.2022.100027.
- Hair, J. F., G. T. M. Hult, C. M. Ringle, M. Sarstedt, N. P. Danks, and S. Ray. 2021. "The SEMinR Package." In *Partial Least Squares Structural Equation Modeling (PLS-SEM) Using R*, 49–74 Cham: Springer. doi:10.1007/978-3-030-80519-7
- Hair, J. F., G. T. M. Hult, C. M. Ringle, M. Sarstedt, and K. O. Thiele. 2017. "Mirror, Mirror on the Wall: A Comparative Evaluation of Composite-Based Structural Equation Modeling Methods." *Journal of the Academy of Marketing Science* 45 (5): 616–632. doi:10.1007/s11747-017-0517-x.
- Hair, J. F., Jr, G. T. M. Hult, C. M. Ringle, and M. Sarstedt. 2021. *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)*. Thousand Oaks, CA: Sage publications.
- Hair, J. F., Jr, G. T. M. Hult, C. M. Ringle, M. Sarstedt, N. P. Danks, and S. Ray. 2021. *Partial Least Squares Structural Equation Modeling (PLS-SEM) Using R: A Workbook*. Switzerland: Springer Nature.
- Hamadneh, J., S. Duleba, and D. Esztergár-Kiss. 2022. "Stakeholder Viewpoints Analysis of the Autonomous Vehicle Industry by Using Multi-Actors Multi-Criteria Analysis." *Transport Policy* 126: 65–84. doi:10.1016/j.tranpol.2022.07.005.
- Hassan, H. M., M. R. Ferguson, S. Razavi, and B. Vrkljan. 2019. "Factors That Influence Older Canadians' Preferences for Using Autonomous Vehicle Technology: A Structural Equation Analysis." *Transportation Research Record: Journal of the Transportation Research Board* 2673 (1): 469–480. doi:10.1177/0361198118822281.
- Haustein, S., and A. Siren. 2014. "Seniors' Unmet Mobility Needs—How Important is a Driving Licence?" *Journal of Transport Geography* 41: 45–52. doi:10.1016/j.jtrangeo.2014.08.001.
- He, W., D. Goodkind, and P. Kowal. 2016. *US Census Bureau, International Population Reports, P95/16-1, an Aging World: 2015*. Washington, DC: US Government Publishing Office.
- Hu, L.-t., and P. M. Bentler. 1998. "Fit Indices in Covariance Structure Modeling: Sensitivity to Underparameterized Model Misspecification." *Psychological Methods* 3 (4): 424–453. doi:10.1037/1082-989X.3.4.424.
- Huang, G., Y.-H. Hung, R. W. Proctor, and B. J. Pitts. 2021. "Non-Chronological Age Factors and Self-Perceived Driving Abilities: A Survey Study of Autonomous Vehicle Acceptance." *Technology, Mind & Society 2021 Conference Proceedings*.
- Hulland, J. 1999. "Use of Partial Least Squares (PLS) in Strategic Management Research: A Review of Four Recent Studies." *Strategic Management Journal* 20 (2): 195–204. doi:10.1002/(SICI)1097-0266(199902)20:2<195::AID-SMJ13>3.0.CO;2-7.
- Husson, F., S. Lê, and J. Pagès. 2011. *Exploratory Multivariate Analysis by Example Using R*. Vol. 15. Boca Raton: CRC Press.
- Kacperski, C., T. Vogel, and F. Kutzner. 2020. "Ambivalence in Stakeholders' Views on Connected and Autonomous Vehicles." In *HCI in Mobility, Transport, and Automotive Systems. Automated Driving and In-Vehicle Experience Design: Second International Conference, MobiTAS 2020, Held as Part of the 22nd HCI International Conference, HCII 2020, Copenhagen, Denmark, July 19–24, 2020, Proceedings, Part I* 22 (pp. 46–57). Springer International Publishing.
- Kadylak, T., S. R. Cotten, and C. Fennell. 2021. "Willingness to Use Automated Vehicles: Results from a Large and Diverse Sample of US Older Adults." *Gerontology & Geriatric Medicine* 7: 2333721420987335. doi:10.1177/2333721420987335.
- Kasai, M., T. Sugawara, J. Takada, K. Kumai, K. Nakamura, and K. Meguro. 2021. "Self-Reported Cognitive Decline Based on the Ascertain Dementia 8 Questionnaire May Be Effective for Detecting Mild Cognitive Impairment and Dementia in the Community: The Wakuya Project." *Dementia and Geriatric Cognitive Disorders Extra* 11 (1): 45–50. doi:10.1159/000514324.
- Kassambara, A. 2017. "Principal Component Methods in R: Practical Guide." *Principal Component Methods in R: Practical Guide*. STHDA. <http://www.sthda.com/english/articles/31-principal-component-methods-in-r-practical-guide/112-pca-principal-component-analysis-essentials/>
- Kelley, C. R. 1969. "What is Adaptive Training?" *Human Factors: The Journal of the Human Factors and Ergonomics Society* 11 (6): 547–556. doi:10.1177/001872086901100602.
- Kempf, C., L. R. Schwarz, A. K. Desai, and M. A. Perkinson. 2016. "Cognitive Impairment, Dementia, and Occupational Therapy Interventions." In edited by Frank Barney K., C. Emerita, and M. A. Perkinson. *Occupational Therapy with Aging Adults*, 245–265. Elsevier.
- Knoefel, F., B. Wallace, R. Goubran, I. Sabra, and S. Marshall. 2019. "Semi-Autonomous Vehicles as a Cognitive Assistive Device for Older Adults." *Geriatrics* 4 (4): 63. doi:10.3390/geriatrics4040063.
- Kono, S., and M. Sato. 2022. "The Potentials of Partial Least Squares Structural Equation Modeling (PLS-SEM) in Leisure Research." *Journal of Leisure Research* 54 (3): 309–329. doi:10.1080/00222216.2022.2066492.
- Lajunen, T., and M. J. Sullman. 2021. "Attitudes toward Four Levels of Self-Driving Technology among Elderly Drivers."

- Frontiers in Psychology* 12: 682973. doi:10.3389/fpsyg.2021.682973.
- Landau, R., G. K. Auslander, S. Werner, N. Shoval, and J. Heinik. 2010. "Families' and Professional Caregivers' Views of Using Advanced Technology to Track People with Dementia." *Qualitative Health Research* 20 (3): 409–419. doi:10.1177/1049732309359171.
- Lang, Y., L. Wei, F. Xu, Y. Zhao, and L.-F. Yu. 2018. "Synthesizing Personalized Training Programs for Improving Driving Habits via Virtual Reality." 2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), Reutlingen, Germany, 297–304.
- Le Roux, B., and H. Rouanet. 2010. *Multiple Correspondence Analysis*. Vol. 163. CA: Sage.
- Lee, J. D., and K. A. See. 2004. "Trust in Automation: Designing for Appropriate Reliance." *Human Factors* 46 (1): 50–80. doi:10.1518/hfes.46.1.50_30392.
- Litman, T. 2021. *Evaluating Transportation Equity: Guidance for Incorporating Distributional Impacts in Transport Planning*. Institute of Transportation Engineers. ITE Journal, 92 (4): 43–49.
- Liu, P., Y. Zhang, and Z. He. 2019. "The Effect of Population Age on the Acceptable Safety of Self-Driving Vehicles." *Reliability Engineering & System Safety* 185: 341–347. doi:10.1016/j.res.2019.01.003.
- Lohmöller, J.-B. 1989. *Latent Variable Path Modeling with Partial Least Squares*. Heidelberg: Springer Science & Business Media.
- Lyman, J. M., G. McGwin, Jr, and R. V. Sims. 2001. "Factors Related to Driving Difficulty and Habits in Older Drivers." *Accident; Analysis and Prevention* 33 (3): 413–421. doi:10.1016/s0001-4575(00)00055-5.
- Maresova, P., S. Tomsone, P. Lameski, J. Madureira, A. Mendes, E. Zdravetski, I. Chorbev, V. Trajkovik, M. Ellen, and K. Rodile. 2018. "Technological Solutions for Older People with Alzheimer's Disease: Review." *Current Alzheimer Research* 15 (10): 975–983. doi:10.2174/1567205015666180427124547.
- May, J. F., and C. L. Baldwin. 2009. "Driver Fatigue: The Importance of Identifying Causal Factors of Fatigue When considering Detection and Countermeasure Technologies." *Transportation Research Part F: Traffic Psychology and Behaviour* 12 (3): 218–224. doi:10.1016/j.trf.2008.11.005.
- Metz, D. H. 2000. "Mobility of Older People and Their Quality of Life." *Transport Policy*, 7 (2): 149–152. doi:10.1016/S0967-070X(00)00004-4.
- Mitra, S. K., Y. Bae, and S. G. Ritchie. 2019. "Use of Ride-Hailing Services among Older Adults in the United States." *Transportation Research Record* 2673 (3): 700–710. doi:10.1177/0361198119835511.
- Monica, M. M., M. Díaz-Santos, and K. Vossel. 2021. "Alzheimer's Association 2021 Facts and Figures Report." Alzheimer's Association.
- Nash, A. 2019. *National Population Projections: 2018-Based*. UK: Office for National Statistics.
- Nasreddine, Z. S., N. A. Phillips, V. Bédirian, S. Charbonneau, V. Whitehead, I. Collin, J. L. Cummings, and H. Chertkow. 2005. "The Montreal Cognitive Assessment, MoCA: A Brief Screening Tool for Mild Cognitive Impairment." *Journal of the American Geriatrics Society* 53 (4): 695–699. doi:10.1111/j.1532-5415.2005.53221.x.
- National Research Council. 2014. *Autonomy Research for Civil Aviation: Toward a New Era of Flight*. Washington, DC : National Academies Press.
- O'Connor, M. L., J. D. Edwards, V. G. Wadley, and M. Crowe. 2010. "Changes in Mobility among Older Adults with Psychometrically Defined Mild Cognitive Impairment." *The Journals of Gerontology-Series B, Psychological Sciences and Social Sciences* 65B (3): 306–316. doi:10.1093/geronb/gbq003.
- Okamoto, S., E. Kobayashi, H. Murayama, J. Liang, T. Fukaya, and S. Shinkai. 2021. "Decomposition of Gender Differences in Cognitive Functioning: National Survey of the Japanese Elderly." *BMC Geriatrics* 21 (1): 38. doi:10.1186/s12877-020-01990-1.
- Parasuraman, R., and V. Riley. 1997. "Humans and Automation: Use, Misuse, Disuse, Abuse." *Human Factors: The Journal of the Human Factors and Ergonomics Society* 39 (2): 230–253. doi:10.1518/001872097778543886.
- Park, J., and S. Han. 2023. "Investigating Older Consumers' Acceptance Factors of Autonomous Vehicles." *Journal of Retailing and Consumer Services* 72: 103241. doi:10.1016/j.jretconser.2022.103241.
- Park, J., M. Zahabi, S. Blanchard, X. Zheng, M. Ory, and M. Benden. 2023. "A Novel Autonomous Vehicle Interface for Older Adults with Cognitive Impairment." *Applied Ergonomics* 113: 104080. doi:10.1016/j.apergo.2023.104080.
- Pedersen, M. M., N. E. Holt, L. Grande, L. A. Kurlinski, M. K. Beauchamp, D. K. Kiely, J. Petersen, S. Leveille, and J. F. Bean. 2014. "Mild Cognitive Impairment Status and Mobility Performance: An Analysis from the Boston RISE Study." *The Journals of Gerontology-Series A, Biological Sciences and Medical Sciences* 69 (12): 1511–1518. doi:10.1093/gerona/glu063.
- Perneckzy, R., C. Pohl, C. Sorg, J. Hartmann, N. Tomic, T. Grimmer, S. Heitele, and A. Kurz. 2006. "Impairment of Activities of Daily Living Requiring Memory or Complex Reasoning as Part of the MCI Syndrome." *International Journal of Geriatric Psychiatry* 21 (2): 158–162. doi:10.1002/gps.1444.
- Rahman, M. M., S. Deb, L. Strawderman, R. Burch, and B. Smith. 2019. "How the Older Population Perceives Self-Driving Vehicles." *Transportation Research Part F: Traffic Psychology and Behaviour* 65: 242–257. doi:10.1016/j.trf.2019.08.002.
- Raithel, S., M. Sarstedt, S. Scharf, and M. Schwaiger. 2012. "On the Value Relevance of Customer Satisfaction. Multiple Drivers and Multiple Markets." *Journal of the Academy of Marketing Science* 40 (4): 509–525. doi:10.1007/s11747-011-0247-4.
- Ranchet, M., M. Tant, A. E. Akinwuntan, J. C. Morgan, and H. Devos. 2017. "Fitness-to-Drive Disagreements in Individuals with Dementia." *The Gerontologist* 57 (5): 833–837. doi:10.1093/geront/gnw119.
- Ravensbergen, L., K. B. Newbold, R. Ganann, and C. Sinding. 2021. "Mobility Work: Older Adults' Experiences Using Public Transportation." *Journal of Transport Geography* 97: 103221. doi:10.1016/j.jtrangeo.2021.103221.
- Ravensbergen, L., M. Van Liefveringe, J. Isabella, Z. Merrina, and A. El-Geneidy. 2022. "Accessibility by Public Transport for Older Adults: A Systematic Review." *Journal of Transport Geography* 103: 103408. doi:10.1016/j.jtrangeo.2022.103408.

- Rodriguez-Sabate, C., I. Morales, A. Sanchez, and M. Rodriguez. 2017. "The Multiple Correspondence Analysis Method and Brain Functional Connectivity: its Application to the Study of the Non-Linear Relationships of Motor Cortex and Basal Ganglia." *Frontiers in Neuroscience* 11: 345. doi:10.3389/fnins.2017.00345.
- SAE. 2018. *Taxonomy and Definitions for Terms Related to Driving Automation Systems for on-Road Motor Vehicles*. Warrendale, PA, USA: SAE International.
- SAE. 2021. "SAE Levels of Driving Automation™ Refined for Clarity and International Audience." <https://www.sae.org/blog/sae-j3016-update>.
- Sallis, J. F., N. Owen, and E. Fisher. 2015. "Ecological Models of Health Behavior." *Health Behavior: Theory, Research, and Practice*, 5: 43–64.
- Salthouse, T. 2010. *Major Issues in Cognitive Aging*. Oxford, UK: Oxford University Press.
- Satariano, W. A., J. M. Guralnik, R. J. Jackson, R. A. Marottoli, E. A. Phelan, and T. R. Prohaska. 2012. "Mobility and Aging: new Directions for Public Health Action." *American Journal of Public Health* 102 (8): 1508–1515. doi:10.2105/AJPH.2011.300631.
- Saxby, D. J., G. Matthews, J. S. Warm, E. M. Hitchcock, and C. Neubauer. 2013. "Active and Passive Fatigue in Simulated Driving: discriminating Styles of Workload Regulation and Their Safety Impacts." *Journal of Experimental Psychology-Applied* 19 (4): 287–300. doi:10.1037/a0034386.
- Shen, S., M. Ramirez, C. J. Hamann, N. Morris, C. Peek-Asa, and M. Zhu. 2021. "The Associations between Older Driver Licensure Laws with Travel and Passenger Behaviors among Adults Aged 65 Years or Older (United States, 2003–2017)." *International Journal of Environmental Research and Public Health* 18 (5): 2251. doi:10.3390/ijerph18052251.
- Shoval, N., H.-W. Wahl, G. Auslander, M. Isaacson, F. Oswald, T. Edry, R. Landau, and J. Heinik. 2011. "Use of the Global Positioning System to Measure the out-of-Home Mobility of Older Adults with Differing Cognitive Functioning." *Ageing and Society* 31 (5): 849–869. doi:10.1017/S0144686X10001455.
- Siegfried, A. L., A. Bayne, L. F. Beck, and K. Freund. 2021. "Older Adult Willingness to Use Fully Autonomous Vehicle (FAV) Ride Sharing." *Geriatrics*, 6 (2): 47. doi:10.3390/geriatrics6020047.
- Solli, H., S. Hvalvik, I. T. Bjørk, and R. Hellesø. 2015. "Characteristics of the Relationship That Develops from Nurse-Caregiver Communication during Telecare." *Journal of Clinical Nursing* 24 (13–14): 1995–2004. doi:10.1111/jocn.12786.
- Soubra, R., A. Chkeir, and J.-L. Novella. 2019. "A Systematic Review of Thirty-One Assessment Tests to Evaluate Mobility in Older Adults." *BioMed Research International* 2019: 1354362–1354317. doi:10.1155/2019/1354362.
- Souders, D., and N. Charness. 2016. "Challenges of Older Drivers' Adoption of Advanced Driver Assistance Systems and Autonomous Vehicles. Human Aspects of IT for the Aged Population." Healthy and Active Aging: Second International Conference, ITAP 2016, Held as Part of HCI International 2016 Toronto, Canada, July 17–22, 2016, Proceedings, Part II 2.
- Souders, D. J., R. Best, and N. Charness. 2017. "Valuation of Active Blind Spot Detection Systems by Younger and Older Adults." *Accident; Analysis and Prevention* 106: 505–514. doi:10.1016/j.aap.2016.08.020.
- St-Onge, M.-P. 2005. "Relationship between Body Composition Changes and Changes in Physical Function and Metabolic Risk Factors in Aging." *Current Opinion in Clinical Nutrition & Metabolic Care* 8 (5): 523–528. doi:10.1097/01.mco.0000171150.49248.14.
- Stanley, J. K., D. A. Hensher, J. R. Stanley, and D. Vella-Brodrick. 2011. "Mobility, Social Exclusion and Well-Being: Exploring the Links." *Transportation Research Part A: Policy and Practice* 45 (8): 789–801. doi:10.1016/j.tra.2011.06.007.
- Stelmach, G. E., and A. Nahom. 1992. "Cognitive-Motor Abilities of the Elderly Driver." *Human Factors* 34 (1): 53–65. doi:10.1177/001872089203400107.
- Thomas, A. K., and A. Gutchess. 2020. *The Cambridge Handbook of Cognitive Aging: A Life Course Perspective*. Cambridge, UK: Cambridge University Press.
- Tolea, M. I., and J. E. Galvin. 2016. "Relationship between Mobility Dysfunction Staging and Global Cognitive Performance." *Alzheimer Disease and Associated Disorders* 30 (3): 230–236. doi:10.1097/WAD.0000000000000136.
- Vaissie, P., A. Monge, and F. Husson. 2015. "Factoshiny: perform Factorial Analysis from FactoMineR with a Shiny Application." In R Package Version. <https://cran.r-project.org/web/packages/Factoshiny/Factoshiny.pdf>.
- Valentine, R. J., M. M. Misic, K. S. Rosengren, J. A. Woods, and E. M. Evans. 2009. "Sex Impacts the Relation between Body Composition and Physical Function in Older Adults." *Menopause* 16 (3): 518–523. doi:10.1097/gme.0b013e31818c931f.
- van Boekel, L. C., E. J. Wouters, B. M. Grimberg, N. J. van der Meer, and K. G. Luijkx. 2019. "Perspectives of Stakeholders on Technology Use in the Care of Community-Living Older Adults with Dementia: A Systematic Literature Review." *Healthcare* 7 (2): 73. doi:10.3390/healthcare7020073.
- Viljanen, A., T. M. Mikkola, M. Rantakokko, E. Portegijs, and T. Rantanen. 2016. "The Association between Transportation and Life-Space Mobility in Community-Dwelling Older People with or without Walking Difficulties." *Journal of Aging and Health* 28 (6): 1038–1054. doi:10.1177/0898264315618919.
- Wadley, V. G., O. Okonkwo, M. Crowe, D. E. Vance, J. M. Elgin, K. K. Ball, and C. Owsley. 2009. "Mild Cognitive Impairment and Everyday Function: An Investigation of Driving Performance." *Journal of Geriatric Psychiatry and Neurology* 22 (2): 87–94. doi:10.1177/0891988708328215.
- Webber, S. C., M. M. Porter, and V. H. Menec. 2010. "Mobility in Older Adults: A Comprehensive Framework." *The Gerontologist* 50 (4): 443–450. doi:10.1093/geront/gnq013.
- Wöbbecking-Sánchez, M., B. Bonete-López, A. S. Cabaco, J. D. Urchaga-Litago, and R. M. Afonso. 2020. "Relationship between Cognitive Reserve and Cognitive Impairment in Autonomous and Institutionalized Older Adults." *International Journal of Environmental Research and Public Health* 17 (16): 5777. doi:10.3390/ijerph17165777.
- Wolf, E. J., K. M. Harrington, S. L. Clark, and M. W. Miller. 2013. "Sample Size Requirements for Structural Equation Models: An Evaluation of Power, Bias, and Solution Propriety." *Educational and Psychological Measurement* 76 (6): 913–934. doi:10.1177/0013164413495237.
- Zahabi, M., P. Machado, C. Pankok, Jr, M. Y. Lau, Y.-F. Liao, J. Hummer, W. Rasdorf, and D. B. Kaber. 2017. "The Role of

- Driver Age in Performance and Attention Allocation Effects of Roadway Sign Count, Format and Familiarity." *Applied Ergonomics* 63: 17–30. doi:[10.1016/j.apergo.2017.04.001](https://doi.org/10.1016/j.apergo.2017.04.001).
- Zahabi, M., J. Park, A. M. A. Razak, and A. D. McDonald. 2020. "Adaptive Driving Simulation-Based Training: framework, Status, and Needs." *Theoretical Issues in Ergonomics Science* 21 (5): 537–561. doi:[10.1080/1463922X.2019.1698673](https://doi.org/10.1080/1463922X.2019.1698673).
- Zahabi, M., A. M. A. Razak, R. K. Mehta, and M. Manser. 2021. "Effect of Advanced Driver-Assistance System Trainings on Driver Workload, Knowledge, and Trust." *Transportation Research Part F: Traffic Psychology and Behaviour* 76: 309–320. doi:[10.1016/j.trf.2020.12.003](https://doi.org/10.1016/j.trf.2020.12.003).
- Zahabi, M., A. M. A. Razak, A. E. Shortz, R. K. Mehta, and M. Manser. 2020. "Evaluating Advanced Driver-Assistance System Trainings Using Driver Performance, Attention Allocation, and Neural Efficiency Measures." *Applied Ergonomics* 84: 103036. doi:[10.1016/j.apergo.2019.103036](https://doi.org/10.1016/j.apergo.2019.103036).
- Zandieh, R., and R. A. Acheampong. 2021. "Mobility and Healthy Ageing in the City: Exploring Opportunities and Challenges of Autonomous Vehicles for Older Adults' Outdoor Mobility." *Cities* 112: 103135. doi:[10.1016/j.cities.2021.103135](https://doi.org/10.1016/j.cities.2021.103135).